

- Business Requirements Document (BRD)**
- Project Title:** Data Integration and Aggregation for Customer and Order Data
- Introduction:**

The purpose of this project is to design and implement a data integration and aggregation solution for customer and order data. The solution will involve reading source CSV data, loading it into Delta tables, performing data cleansing and transformation, and creating aggregated reports.

- Business Requirements:**

1. **Data Ingestion**

- Read source CSV data from specified volumes:
``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and
``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``
- Load data into Delta tables for customer and order data

2. **Data Schema**

- Customer table schema: ``CustId`, `Name`, `EmailId`, `Region``
- Order table schema: ``OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId``

3. **Data Cleansing**

- Remove "Null"/null and duplicate records from both customer and order tables

4. **Data Transformation**

- Create an "ordersummary" table with schema: ``CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date``
- Join customer and order data using the "CustId" field and load the data into an SCD Type 2 table ("ordersummary")

5. **SCD Type 2 Implementation**

- Update the SCD Type 2 table ("ordersummary") whenever there is a change in the customer table
- Old records should be made inactive and new records should be made active, with corresponding changes to ``StartDate`` and ``EndDate``

6. **Aggregated Reporting**

- Create a new table "customeraggregatespend" with columns "Name", "TotalAmount", and "Date"
- Aggregate the "TotalAmount" column from the "ordersummary" table and group by "Name" and "Date" columns
- Load the aggregated data into the "customeraggregatespend" table

- Functional Requirements:**

1. The system should be able to read CSV data from specified volumes.
2. The system should be able to load data into Delta tables.
3. The system should perform data cleansing (remove "Null"/null and duplicate records).
4. The system should create and update the "ordersummary" table with the required schema.
5. The system should implement SCD Type 2 logic for the "ordersummary" table.
6. The system should create and update the "customeraggregatespend" table with aggregated data.

- Non-Functional Requirements:**

1. The system should ensure data consistency and integrity.
2. The system should be scalable to handle large volumes of data.
3. The system should be able to handle changes in the customer table and update the "ordersummary" table accordingly.

- Assumptions and Dependencies:**

1. The source CSV data is available in the specified volumes.
2. The Delta tables are created and managed by the system.
3. The system has the necessary permissions and access to read and write data.

- Acceptance Criteria:**

1. The system successfully reads CSV data and loads it into Delta tables.
2. The system performs data cleansing and removes "Null"/null and duplicate records.
3. The system creates and updates the "ordersummary" table with the required schema.
4. The system implements SCD Type 2 logic for the "ordersummary" table.
5. The system creates and updates the "customeraggregatespend" table with aggregated data.

- Data Dictionary:**

- Customer table: `CustId`, `Name`, `EmailId`, `Region`
- Order table: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`
- Ordersummary table: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`
- Customeraggregatespend table: `Name`, `TotalAmount`, `Date`

This BRD provides a comprehensive overview of the business requirements, functional requirements, and non-functional requirements for the data integration and aggregation project. It serves as a foundation for further design, development, and testing activities.